



Jungho Kim

Ph.D. Student

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Research Interests

- Safety-Oriented End-to-End Autonomous Driving
- 3D Scene Understanding and Occupancy Prediction
- Motion Prediction and Planning
- Autonomous Driving Foundation Models

Education

Ph.D. in Interdisciplinary Program in Artificial Intelligence

Mar 2025 – Present

Seoul National University, Seoul, South Korea

Advisor: [Prof. Jun Won Choi](#)

M.S. in Artificial Intelligence

Mar 2022 – Feb 2025

Hanyang University, Seoul, South Korea

Advisor: [Prof. Jun Won Choi](#)

B.S. in Automotive Engineering

Mar 2015 – Feb 2022

Kookmin University, Seoul, South Korea

Minor: [Computer Science](#)

Publications (* : Equal Contributions)

SafeDrive: Fine-Grained Safety Reasoning for End-to-End Driving in a Sparse World

Jungho Kim, Jiyong Oh, Seunghoon Yu, Hongjae Shin, Donghyuk Kwak, and Jun Won Choi

[The IEEE/CVF Conference on Computer Vision and Pattern Recognition \(CVPR Highlight\)](#), 2026.

STONE Dataset: Scalable 3D Traversability Dataset with Multi-sensor Surround-View for Robot Navigation in Off-road Environments

Konyul Park*, Daehun Kim*, Jiyong Oh, Seunghoon Yu, Junseo Park, Jaehyun Park, Hongjae Shin, Hyungchan Cho, **Jungho Kim**, and Jun Won Choi

[IEEE International Conference on Robotics & Automation \(ICRA\)](#), 2026.

OnlineBEV: Recurrent Temporal Fusion in Bird's Eye View Representations for Multi-Camera 3D Perception

Junho Koh*, Youngwoo Lee*, **Jungho Kim**, Dongyoung Lee, and Jun Won Choi

[IEEE Transactions on Intelligent Transportation Systems \(T-ITS\)](#), 2025.

ProtoOcc: Accurate, Efficient 3D Occupancy Prediction Using Dual Branch Encoder-Prototype Query Decoder

Jungho Kim*, Changwon Kang*, Dongyoung Lee*, Sehwan Choi, and Jun Won Choi

[AAAI Conference on Artificial Intelligence \(AAAI\), 2025.](#)

Mask2Map: Vectorized HD Map Construction Using Bird's Eye View Segmentation Masks

Sehwan Choi*, Jungho Kim*, Hongjae Shin, and Jun Won Choi

[European Conference on Computer Vision \(ECCV Oral\), 2024.](#)

SiT Dataset: Socially Interactive Pedestrian Trajectory Dataset for Social Navigation Robots

Jong Wook Bae*, Jungho Kim*, Junyong Yun*, Changwon Kang*, Jeongseon Choi, Chanhyeok Kim, Junho Lee, Jungwook Choi, Chanhyeok Kim, Junho Lee, Jungwook Choi, and Jun Won Choi

[Neural Information Processing Systems \(NeurIPS\), 2023.](#)

R-Pred: Two-Stage Motion Prediction Via Tube-Query Attention-Based Trajectory Refinement

Sehwan Choi, Jungho Kim, Junyong Yun, and Jun Won Choi

[IEEE/CVF International Conference on Computer Vision \(ICCV\), 2023.](#)

CarPLAN: Context-Adaptive and Robust Planning with Dynamic Scene Awareness for Autonomous Driving

Junyong Yun*, Jungho Kim*, ByungHyun Lee, Dongyoung Lee, Sehwan Choi, Seunghyeop Nam, Kichun Jo, and Jun Won Choi

[Submitted, 2025.](#)

Competition

NAVSIM v2 End-to-End Driving Challenge 2025 (CVPR)

Team SNU

[Travel Grant Recipient](#)

Research Projects

Development of Reference Data and Core Technologies for End-to-End Autonomous Driving

[Government-Funded National Research Project](#) · 🏢 Jul. 2025 – Dec. 2028

- Develop reference datasets and end-to-end autonomous driving models for real-vehicle deployment.
- Develop perception components that support downstream prediction, planning, and control in an integrated driving system.

Large-Model-Based Global Map Generation Technology Development

[Hyundai Motor Group](#) · 🏢 Dec. 2025 – Nov. 2026

- Develop global map generation methods leveraging large models and diverse driving data.
- Improve map consistency and scalability across long driving trajectories and large-scale environments.

Exploratory Development of Planning Networks for Autonomous Driving

[Hyundai Mobis](#) · 🏢 Jan. 2026 – Dec. 2026

- Investigate learning-based planning network architectures for autonomous driving.
- Develop training objectives and evaluation methods for safe planning in dynamic traffic scenarios.

Mobility Technologies for Rapid and Safe Autonomous Driving in Unstructured Terrain

[Government-Funded National Research Project](#) · 🏢 Apr. 2024 – Dec. 2025

- Develop environment-recognition technologies for unstructured off-road driving scenarios.
- Develop safety-aware mobility technologies for robust autonomous navigation on diverse terrains.

Deep Network Development for Camera-Based Multi-Purpose Object Detection

[Hyundai Motor Group](#) · 🏢 Jun. 2023 – May 2024

- Develop deep neural networks for camera-based multi-purpose object detection.
- Improve robust recognition of surrounding road users and objects under diverse driving conditions.

Patent

Method for Occupancy State Prediction of Semantic Objects Based on Multi-View Images for Surrounding-Environment Recognition, and Computer Device Therefor

Republic of Korea Patent Application No. 10-2024-0182281
Jun Won Choi, Sehwan Choi, and **Jungho Kim**.

Method for Generating a Voxel Feature Map Based on a Bird’s-Eye-View Feature Map, Vehicle, and Computer-Readable Storage Medium Storing Instructions for Performing the Method

Republic of Korea Patent Application No. 10-2025-0109147
Jun Won Choi and **Jungho Kim**.